

Transport through a ferromagnet-bilayer-graphene-superconductor junction

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(Dated: September 5, 2026)

I. MODEL AND FORMULATION

The system is illustrated in Fig. 1. It is composed by two layers, labeled by $j = 1, 2$, coupled in two geometries referred as $1 \rightarrow 1$ and $1 \rightarrow 2$. These correspond to the Fig. 1 (a) and Fig. 1 (b), respectively. The transversal and longitudinal direction are indexed by n and m integers, respectively. Using this convention, we notice that the graphene-bilayer is located in the range $1 \leq m \leq N$, for both geometries. For $m \leq 0$, the graphene monolayer is assumed to be ferromagnetic, while for $m \geq N + 1$, there is the s -wave superconductor monolayer. The main difference between both geometries regards the coupling between the graphene-bilayer and the superconductor monolayer at its right side. In fact, for $1 \rightarrow 1$ geometry, the superconductor is coupled to the lower monolayer while in the geometry $1 \rightarrow 2$, the superconductor monolayer is coupled to the upper layer. This is more clearly visualized in Fig. 1(d), where a block diagram of the system is shown. We also point out that for both geometries we have assumed a Bernal stacking, as shown in Fig. 1(c), where an atom of A -atom of the lower monolayer is coupled to an B atom of the upper monolayer.

A. The Model

Next, we define the Hamiltonian for each section of the system illustrated in Fig. 1.

1. Coupling Hamiltonian

By assuming a nearest-neighbour approximation, each A site is connected to three B sites. As a result, the intralayer Hamiltonian is

$$\hat{H}_t = -t \sum_{\substack{j,\sigma \\ m,n}} \hat{a}_{jmn\sigma}^\dagger \left(\hat{b}_{jmn\sigma} + \hat{b}_{j,m+1,n,\sigma} + \hat{b}_{j,m,n+1,\sigma} \right) + \text{H.c.}, \quad (1)$$

where we have also added the indexes $j = (1, 2)$ that identifies the lower and upper monolayers, respectively, and the index $\sigma = (\uparrow, \downarrow)$, in order to identify the electron spin, since the ferromagnet polarization breaks the spin symmetry.

In order to account for the coupling between lower and upper monolayer, we define the Hamiltonian $\hat{H}_\perp$ whose form is given by:

$$\hat{H}_\perp = -t_\perp \sum_{m=1}^N \sum_{n,\sigma} \left(\hat{a}_{1mn\sigma}^\dagger \hat{b}_{2mn\sigma} + \text{H.c.} \right), \quad (2)$$

since we are using AB stacking, the dimer bond connects A_1 to B_2 . Notice that it exists only in range $0 \leq m \leq N$ where the bilayer is defined.

2. Chemical potentials, gate potentials, and exchange field

In order to take the different physical correlations at each section of the system, we define the on-site Hamiltonian as follows:

$$\hat{H}_{\text{on}} = \sum_{j,m,n,\alpha,\sigma} [V_j(m) - \mu(m) - s_\sigma h(m)] \hat{n}_{jmn\alpha\sigma}, \quad (3)$$

where we have defined the number operator,

$$\hat{n}_{jmn\alpha\sigma} = \hat{c}_{jmn\alpha\sigma}^\dagger \hat{c}_{jmn\alpha\sigma}. \quad (4)$$

In Eq. (3), $s_\uparrow = +1$, $s_\downarrow = -1$ and the additional index α identifies which monolayer section is being considered, i.e., ferromagnetic, central bilayer, and right superconductor monolayer. In this way, $\alpha = F, C, S$.

The chemical potential is piecewise constant,

$$\mu(m) = \begin{cases} \mu_F, & m \leq 0, \\ \mu_C, & 1 \leq m \leq N, \\ \mu_S, & m \geq N + 1, \end{cases} \quad (5)$$

and the collinear exchange field is

$$h(m) = \begin{cases} h, & m \leq 0, \\ 0, & m \geq 1. \end{cases} \quad (6)$$

The central layer potentials V_1, V_2 account for a gate bias applied to each monolayer.

3. Onsite spin-singlet pairing

The superconducting lead contains an onsite pair potential,

$$\hat{H}_\Delta = \sum_{j,m,n,\alpha} \left[\Delta(m) \hat{c}_{jmn\alpha\uparrow}^\dagger \hat{c}_{jmn\alpha\downarrow}^\dagger + \Delta^*(m) \hat{c}_{jmn\alpha\downarrow} \hat{c}_{jmn\alpha\uparrow} \right], \quad (7)$$

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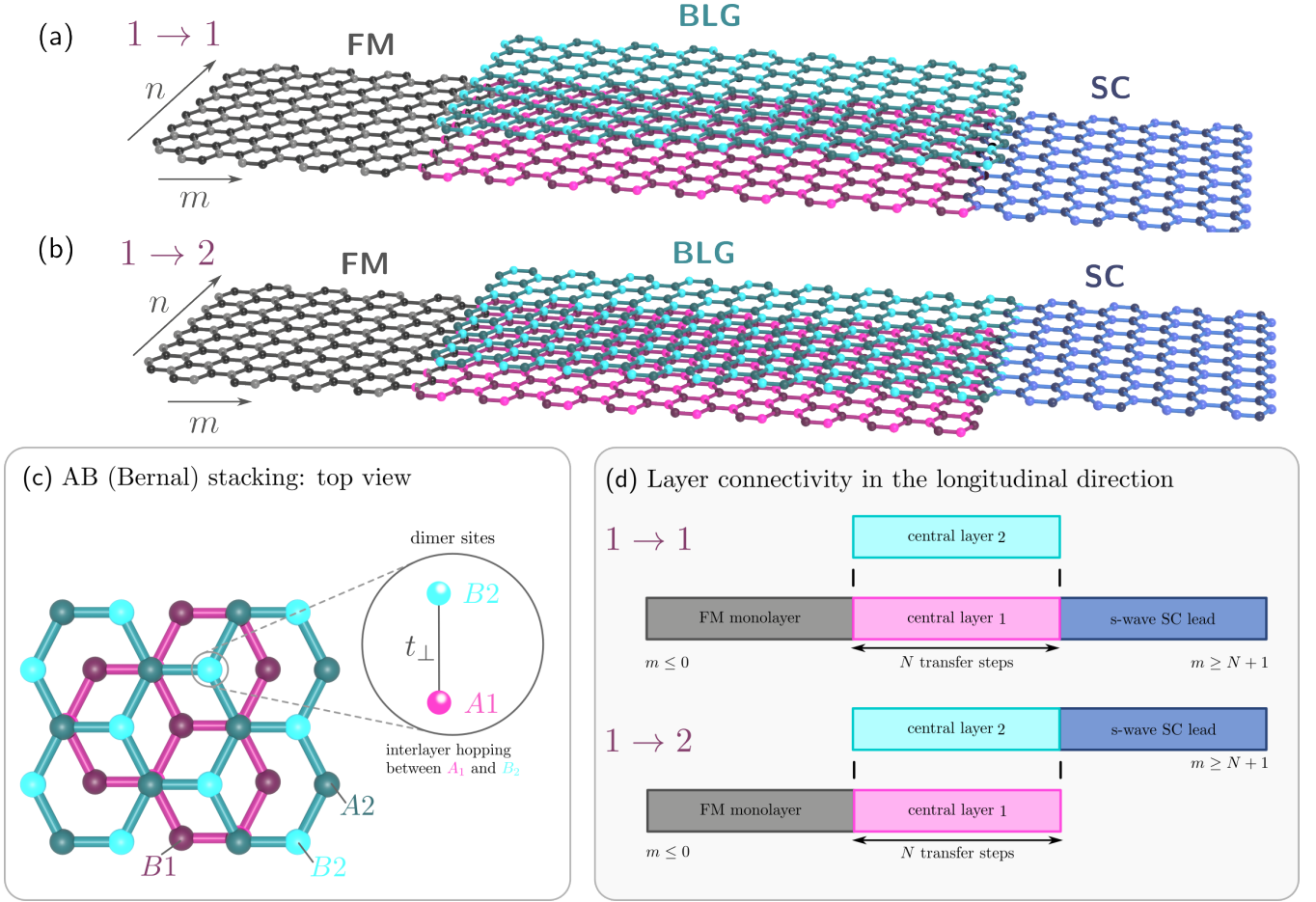


Figure 1. Geometry, lattice structure, and longitudinal connectivity of the ferromagnet–bilayer-graphene–superconductor (FM–BLG–SC) junction. (a) Atomic representation of the $1 \rightarrow 1$ geometry, in which the ferromagnetic (FM) and superconducting (SC) monolayer graphene leads are connected to the same layer of the finite bilayer graphene (BLG) region. (b) Corresponding $1 \rightarrow 2$ geometry, in which the FM and SC leads are connected to opposite graphene layers. The longitudinal and transverse lattice directions are denoted by m and n , respectively. Gray, magenta, cyan, and indigo identify the FM monolayer, central layer 1, central layer 2, and SC monolayer, respectively. (c) Top view of the AB (Bernal) stacking in the central bilayer. Different shades within each layer distinguish the A and B sublattices. The enlarged detail shows the vertically registered A_1 and B_2 dimer sites, coupled by the interlayer hopping $t_{\perp}$; the A_1 site is hidden beneath B_2 in the top projection. The non-dimer sites B_1 and A_2 are also indicated. (d) Longitudinal layer-connectivity representation of the $1 \rightarrow 1$ and $1 \rightarrow 2$ junctions. The FM lead occupies $m \leq 0$, the finite BLG region extends over N transfer steps, $1 \leq m \leq N$, and the SC lead occupies $m \geq N + 1$. In the $1 \rightarrow 1$ geometry the SC lead continues from central layer 1, whereas in the $1 \rightarrow 2$ geometry it continues from central layer 2.

with

$$\Delta(m) = \begin{cases} 0, & m \leq N, \\ \Delta_0 e^{i\varphi}, & m \geq N + 1. \end{cases} \quad (8)$$

Because pairing is onsite, it connects electron and hole amplitudes on the same layer and sublattice. There is no pairing during propagation through the central bilayer.

4. Complete Hamiltonian

By combining Eqs. (1), (2), (3), (7) we have the following complete Hamiltonian for the whole system that

constitutes the microscopic model of the system illustrated in Fig. 1:

$$\hat{H} = \hat{H}_t + \hat{H}_{\perp} + \hat{H}_{\text{on}} + \hat{H}_{\Delta}. \quad (9)$$

In Eq. (9), no interface barrier is added phenomenologically. Contact mismatch arises from the actual change in layer number, exchange field, chemical potential, and pairing.

B. Transverse Fourier transformation and Bogoliubov-de Gennes (BdG) equations

We assume translational invariance along the transversal direction, in this way, we can perform a Fourier trans-

formation along the y -direction. By assuming N_y transverse cells with periodic boundary conditions, we write:

$$\begin{aligned}\hat{a}_{jmn\sigma} &= \frac{1}{\sqrt{N_y}} \sum_K e^{iKn} \hat{a}_{jmK\sigma}, \\ \hat{b}_{jmn\sigma} &= \frac{1}{\sqrt{N_y}} \sum_K e^{iKn} \hat{b}_{jmK\sigma},\end{aligned}$$

with K being the transverse momentum. By substituting these expressions into Eq. (9), we obtain a set of

independent equations. In particular, Eq. (1) becomes:

$$\hat{H}_t(K) = - \sum_{j,m,\sigma} \left[\eta \hat{a}_{jmK\sigma}^\dagger \hat{b}_{jmK\sigma} + \xi \hat{a}_{jmK\sigma}^\dagger \hat{b}_{j,m+1,K,\sigma} + \text{H.c.} \right], \quad (10)$$

with

$$\eta(K) = t(1 + e^{iK}), \quad \xi = t, \quad (11)$$

where the symbol η denotes the effective intracell hopping and ξ the longitudinal intercell hopping. Note that ξ is real, a fact used repeatedly below; η is complex, and its phase is not removable because it is fixed relative to ξ .

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